

Chapter 10 Answer Key

Introduction to Statistics — MATH& 146

Section 10.1 Exercises (Scatterplots)

10.1.1. *Describe direction, form, and likely strength of the association you would expect between each pair.*

(a) Hours of sleep and reaction time. Negative association: more sleep is associated with faster (lower) reaction times. Form is roughly linear over typical sleep ranges, possibly leveling off at very high sleep amounts. Strength is moderate; sleep is one of several factors that affect reaction time (caffeine, age, motivation, time of day).

(b) Car age and resale value. Negative association: older cars are worth less. Form is mildly curved rather than perfectly linear; cars lose value quickly in the first few years and more slowly afterward. Strength is strong.

(c) Shoe size and exam score. No real association in an adult sample; there is no reason these variables should be related. Form is essentially a featureless cloud. Strength is near zero. (In a sample of children, both variables are tied to age, which would produce an apparent positive association; this is a lurking-variable artifact, not a genuine relationship.)

10.1.2. *Why should you always look at the scatterplot before computing any numerical summary of association?*

Numerical summaries can be misleading when used in isolation. The correlation coefficient r , in particular, can fail to detect important features:

- A strong curved relationship can give r near zero. The variables are tightly related, but not linearly.
- One or two outliers can dramatically inflate or deflate r . The reported number describes the outliers as much as the bulk of the data.
- Two clusters of points can produce a misleading correlation that describes neither cluster well.

The classic illustration is Anscombe's quartet: four datasets with identical means, standard deviations, correlation, and regression line, but four completely different scatterplot shapes. The scatterplot is the only way to see these structural features before summarizing them with numbers.

Section 10.2 Exercises (Correlation)

10.2.1. *Without computing, estimate r for each scatterplot in Exercise 10.1.1.*

These are estimates of what the correlation *should* look like for the variables described:

- Sleep and reaction time: $r \approx -0.5$ to -0.7 (moderate negative).
- Car age and resale value: $r \approx -0.85$ to -0.95 (strong negative; the slight curvature would prevent r from reaching -1 even with perfect data).
- Shoe size and exam score: $r \approx 0$ (no linear association).

10.2.2. *If we re-label the axes of a scatterplot (swap x and y), does r change?*

No. The correlation coefficient is symmetric in its two variables:

$$r_{xy} = r_{yx}.$$

The formula contains the product $(x_i - \bar{x})(y_i - \bar{y})$ divided by $s_x s_y$, and multiplication is commutative, so swapping the labels does not change any term in the sum. This is one of the properties that distinguishes r from the regression line, which *does* change when you swap axes (the slope of y on x is different from the slope of x on y unless $r = \pm 1$).

10.2.3. *A dataset has $r = 0.1$. A student says “There is essentially no relationship between these variables.” Is this necessarily true?*

Not necessarily. $r = 0.1$ tells us there is no strong *linear* relationship, but the variables could still be strongly associated in a non-linear way. Consider, for example, $y = x^2$ for x values symmetric around zero: the relationship is perfect and deterministic, yet $r = 0$ because the positive and negative contributions to the correlation sum cancel out.

What the student is missing is the word “linear.” The correct statement is: “There is essentially no linear relationship between these variables.” Whether there is a curved relationship can only be answered by looking at the scatterplot.

Section 10.3 Exercises (Regression Line)

10.3.1. *For the heart rate example, predict resting heart rate for someone consuming 30 grams of sugar daily. Is this within the data range?*

The line is $\hat{y} = 52.4 + 0.480x$. At $x = 30$:

$$\hat{y} = 52.4 + 0.480 \cdot 30 = 52.4 + 14.4 = 66.8 \text{ bpm.}$$

The data summary gives $\bar{x} = 45$ and $s_x = 12$, so $x = 30$ is $(30 - 45)/12 = -1.25$ standard deviations below the mean. This is comfortably within the typical range of the observed data, so the prediction is interpolation (safe), not extrapolation.

- 10.3.2.** *If $r = 0$ in a dataset, what are b and the regression line? What does this mean about using x to predict y ?*

The slope formula gives

$$b = r \cdot \frac{s_y}{s_x} = 0 \cdot \frac{s_y}{s_x} = 0.$$

The intercept is $a = \bar{y} - b\bar{x} = \bar{y}$, so the regression line is

$$\hat{y} = \bar{y}$$

for every x . The line is horizontal, sitting at the height of the response mean.

Practically, this says: x contributes nothing to predicting y . The best linear prediction we can make is the mean of y , regardless of the value of x . Knowing x does not improve the prediction at all.

- 10.3.3.** *A regression is fit to data with $\bar{x} = 10$, $\bar{y} = 50$. The slope is $b = 3$. Find the intercept a . What point must the line pass through?*

$$a = \bar{y} - b\bar{x} = 50 - 3 \cdot 10 = 20.$$

The line is $\hat{y} = 20 + 3x$.

The line passes through the point $(\bar{x}, \bar{y}) = (10, 50)$. Every least-squares regression line passes through the point of means; this is a direct consequence of the intercept formula $a = \bar{y} - b\bar{x}$, which rearranges to $\bar{y} = a + b\bar{x}$.

Section 10.4 Exercises (Coefficient of Determination)

- 10.4.1.** *House size (x , sq. ft.) and sale price (y , dollars) have $r = 0.88$.*

(a) $r^2 = 0.88^2 = 0.7744$.

(b) About 77.4% of the variability in sale price is explained by house size.

(c) The remaining 22.6% is explained by factors not captured in the model: neighborhood and location, age and condition of the house, number of bedrooms and bathrooms, lot size, school district, market timing, and so on. A model that included some of these variables would have a higher r^2 .

- 10.4.2.** *Model A has $r^2 = 0.25$, Model B has $r^2 = 0.81$. Which model makes more accurate predictions?*

Model B. It explains 81% of the variation in y , compared to only 25% for Model A. A higher r^2 means residuals are smaller on average, so predictions cluster more tightly around the actual y values. Equivalently, Model A leaves 75% of the variation in y unexplained, while Model B leaves only 19% unexplained.

10.4.3. *If $r^2 = 1$, what does the scatterplot look like? What are all the residuals?*

The scatterplot is a perfect straight line: every point (x_i, y_i) lies exactly on the regression line $\hat{y} = a + bx$. Every residual is exactly zero:

$$e_i = y_i - \hat{y}_i = 0 \quad \text{for all } i.$$

The line predicts y from x with no error.

Section 10.5 Exercises (Residual Analysis)

10.5.1. *A residual plot fans out as x increases. Which condition is violated? What does this mean for predictions at large x ?*

The **equal-spread** condition is violated. The vertical spread of the residuals should be roughly constant across all x values; a fan shape means residuals are small for low x but large for high x , so the model's prediction error grows with x .

For predictions, this has two consequences. The point predictions $\hat{y}$ themselves may still be roughly unbiased, but the *uncertainty* around those predictions is much larger at high x than the standard regression formulas suggest. Confidence intervals and prediction intervals computed with the usual formulas will be too narrow at high x and possibly too wide at low x ; they understate how much we should trust predictions at small x and overstate it at large x .

10.5.2. *After fitting a regression, one point has a residual of -12.3 while all other residuals are between -2 and $+2$. What would you do?*

That point is an outlier. The first step is to investigate, not delete:

- (1) Check the original data for a recording or transcription error. A typo is the single most common cause of an extreme residual.
- (2) If the data are correct, examine the case. Is there something unusual about that observation (different population, different measurement method, special circumstances)?
- (3) Check whether the point is also high in leverage (an extreme x value); a high-leverage outlier can substantially change the regression line.
- (4) Refit the regression with and without the point and compare the two lines.
- (5) Report both versions, or report the analysis with the outlier removed plus a clear note about its removal and the reason.

Deleting outliers silently is bad practice. The right response is transparency: document the point, explain what you did, and let the reader see the effect of the decision.

Chapter 10 Review

1. *Physical therapy*: $\bar{x} = 6$, $s_x = 2.5$, $\bar{y} = 40$, $s_y = 12$, $r = 0.85$.

(a) Regression line:

$$b = r \cdot \frac{s_y}{s_x} = 0.85 \cdot \frac{12}{2.5} = 0.85 \cdot 4.8 = 4.08,$$
$$a = \bar{y} - b\bar{x} = 40 - 4.08 \cdot 6 = 40 - 24.48 = 15.52.$$

$$\hat{y} = 15.52 + 4.08x$$

(b) Prediction at $x = 8$ weeks:

$$\hat{y} = 15.52 + 4.08 \cdot 8 = 15.52 + 32.64 = 48.16.$$

The patient is predicted to have an improvement score of about 48.2 after 8 weeks of therapy.

(c) $r^2 = 0.85^2 = 0.7225$. About 72.3% of the variability in patient improvement scores is explained by the number of weeks of therapy. The remaining 27.7% comes from other sources: differences in baseline severity, age, adherence to the home exercise plan, comorbidities, and so on.

(d) A curved residual plot violates the linearity condition. Even though $r^2 = 0.72$ is reasonably high, a straight line is the wrong shape for these data; a curved model (quadratic, logarithmic, or saturating) would likely fit better. Predictions from the linear model will be systematically biased: too low in some regions and too high in others. The high r^2 is misleading, because r and r^2 measure linear fit, not best-possible fit.

2. *Cities*: average summer temperature x ($^{\circ}\text{F}$) and per-capita ice cream sales y (dollars). Line: $\hat{y} = -120 + 4.5x$ with $r^2 = 0.61$.

(a) Slope interpretation. For each additional degree Fahrenheit in average summer temperature, predicted annual per-capita ice cream sales increase by \$4.50. In other words, cities that are 10°F warmer on average have about \$45 more per-capita ice cream sales per year, according to the model.

(b) r^2 interpretation. About 61% of the variability in per-capita ice cream sales across these 50 cities is explained by the linear relationship with summer temperature. The remaining 39% reflects other factors: income, demographics, local food culture, competition from frozen-yogurt and gelato shops, and so on.

(c) Predicting at $x = 110^{\circ}\text{F}$ is *not appropriate*. The warmest city in the dataset is 95°F , so 110°F is 15°F beyond any observed data point. This is extrapolation: the linear pattern that holds within the data range may not continue at extreme temperatures. Sales might level off in very hot cities (people may seek air-conditioned indoor activities instead of ice cream) or rise even faster. There is no way to know from the data, because we have no data points at 110°F .

(d) The journalist has reversed the causal direction. The data show that hot summer weather is associated with higher ice cream sales, not that eating ice cream changes summer temperature. Temperature is the explanatory variable; sales are the response. Even if the data were experimental, eating ice cream tends to lower body temperature rather than raise it, so the claim is biologically backwards as well. This is a textbook example of confusing correlation with causation, compounded by getting the causal direction wrong.

3. *Three scatterplots:* $r = -0.92$, $r = 0.45$, $r = 0.01$.

(a) Matching:

- $r = -0.92$: strong negative linear relationship.
- $r = 0.45$: weak (or moderate) positive linear relationship.
- $r = 0.01$: no linear relationship.

(b) Coefficients of determination:

- $r = -0.92$: $r^2 = 0.8464 \approx 84.6\%$ of variability explained.
- $r = 0.45$: $r^2 = 0.2025 \approx 20.3\%$ of variability explained.
- $r = 0.01$: $r^2 = 0.0001 \approx 0.01\%$ of variability explained (essentially zero).

(c) Yes, x and y could still be strongly associated in a non-linear way even with $r = 0.01$. Correlation measures only the linear component of association. A perfect quadratic relationship like $y = x^2$ on symmetric x values gives $r = 0$ but is as strong an association as you can have (one variable determines the other entirely). Looking at the scatterplot is the only way to detect this.

4. *Mutual information from correlation:* $I(X; Y) = -\frac{1}{2} \log_2(1 - r^2)$.

(a) Computing I for each value of r :

r	r^2	$1 - r^2$	$I(X; Y)$ (bits)
0	0	1	$-\frac{1}{2} \log_2(1) = 0$
0.5	0.25	0.75	$-\frac{1}{2} \log_2(0.75) \approx 0.208$
0.9	0.81	0.19	$-\frac{1}{2} \log_2(0.19) \approx 1.198$
0.99	0.9801	0.0199	$-\frac{1}{2} \log_2(0.0199) \approx 2.826$

Notice how slowly I grows at first. A correlation of 0.5 only buys about 0.21 bits of information; you have to push r all the way to 0.99 to get to roughly 2.8 bits.

(b) As $r \rightarrow 1$, we have $1 - r^2 \rightarrow 0$, so $\log_2(1 - r^2) \rightarrow -\infty$ and $I \rightarrow +\infty$. The mutual information diverges. Interpretation: with a perfect linear relationship, knowing X exactly determines Y , so all uncertainty about Y collapses; the amount of information X carries about Y is unlimited. (The infinity is a feature of continuous variables with a deterministic relationship; for discrete variables it would saturate at the entropy of Y .)

(c) The common idea. Chapter 9 defined mutual information for categorical variables as the amount of uncertainty about one variable that is removed by learning the other. The formula $I = -\frac{1}{2} \log_2(1 - r^2)$ is the special case of the same concept for two jointly normal quantitative variables. In both cases, mutual information answers the same question: **how much does knowing one variable reduce my uncertainty about the other?** When the variables are independent, $I = 0$; when one determines the other, I is maximal.

This is why r^2 and mutual information play parallel roles in the course. The categorical and quantitative settings use different languages, but they ask the same underlying question.

5. *The final question: trace “what does one variable tell you about another?” from Chapter 1 to Chapter 10.*

The course can be read as a single extended answer to one question: **what does one variable tell you about another?** Chapters 1 through 3 began with just one variable, asking where it is centered and how much it varies; the **mean** and **standard deviation** are the first tools for describing a single distribution. Chapter 4’s **z-scores** stripped away units so that observations from different distributions could be compared on the same scale, which made the entire inferential framework possible. Chapters 6 and 7 sharpened the question into testable claims with **hypothesis tests**, asking whether one variable’s mean or proportion equals a hypothesized value or differs between two groups. Chapter 8 extended this to count data with **chi-squared tests**, which detect association between two categorical variables, and Chapter 9 quantified the strength of such categorical association with **mutual information**. Chapter 10 finally placed two quantitative variables side by side: **correlation** r measures the direction and strength of their linear association, the **regression line** provides the best linear prediction of one from the other, and r^2 reports the fraction of variability accounted for. The identity $I(X; Y) = -\frac{1}{2} \log_2(1 - r^2)$ ties Chapter 10 back to Chapter 9, revealing that the categorical and quantitative settings are two dialects of the same underlying idea: *statistics is the systematic study of how much one variable tells you about another.*